Feature-Grouped Network With Spectral–Spatial Connected Attention for Hyperspectral Image Classification

Wenhui Guo, Hailiang Ye, and Feilong Cao

***Abstract*— The use of deep learning methods in hyperspectral image (HSI) classification has been a promising approach due to its powerful ability to automatically extract features in recent years. This article proposes a novel deep framework for HSI classification problems, referred to as feature-grouped network based on spectral–spatial connected attention mechanism (FG-SSCA). Different from the existing deep learning methods, the proposed framework integrates the spectral attention module and spatial attention module continuously from the raw HSI input, which is embedded into convolutional neural networks and could enhance the distinguishing ability of spectral bands and learn the spatial relevance between the neighboring pixels together. Meanwhile, the generating feature maps are sliced into a series of small groups in sequence along the direction of spectral bands and each group sequentially extracts spatial–spectral features through multiple spectral and spatial residual blocks. This feature-grouped strategy could fully utilize the redundancy and difference of bands and obtain more available and valuable information. The proposed FG-SSCA method could greatly improve generalization performance and make tremendous successes in HSI classification. Experimental results on several HSI benchmark data sets verify the effectiveness and superiority of the proposed method in comparison with the state-of-the-art approaches for HSI classification.**

***Index Terms*— Deep learning, features grouped, hyperspectral image (HSI) classification, spatial attention, spectral attention.**

I. Introduction

HYPERSPECTRAL images (HSIs) have received considerable attention in the field of remote sensing recently, which include a variety of applications such as urban planning, agriculture, forestry, geography, and military [1], [2]. In contrast to conventional panchromatic and multispectral images, HSI contains hundreds of continuous spectral bands and vast amounts of spectral and spatial information, which are beneficial to distinguishing the targets with different materials. However, the HSI classification is still a challenging task in the remote sensing area due to the inherently high-dimensional structures and abundant information [3], [4]. Accordingly, how to obtain the most discriminative feature from raw data is one of the crucial issues in HSI classification.

Over the past decades, there have been a lot of studies on HSI classification. The classical methods include the kernel-based method [5], [6], random forests [7], manifold ranking method [8], and sparse representation method [9]–[11]. However, these approaches are mainly based on the handcrafted extraction of features, such as independent component analysis [12] and principal component analysis (PCA) [13]. Once the manually extracted features are not appropriate, it would enormously degenerate the classification performance of HSI.

Recently, the rapid development of deep learning, especially the emergence of convolutional neural networks (CNNs), has brought novel technologies to HSI classification due to the extremely powerful ability of automatically captured hierarchical and nonlinear features [14]–[16]. Up to now, many works based on deep learning have been done in the community of HSI classification. Chen *et al.* [17] first applied the deep learning method to HSI classification. Since then, there has been a lot of work on the use of CNNs for HSI classification [18], [19]. For example, Luo *et al.* [20] built a CNN architecture with 1-D manifold embedding (1-D CNN) to extract spectral–spatial feature; Yu *et al.* [21] proposed a feature learning approach based on CNN for HSI classification (referred to as 2-D CNN for avoiding confusion), which learned features from HSIs input data directly; Yang *et al.* [22] learned the deep joint spectral–spatial features with two branches of CNN; and Li *et al.* [23] put forward a 3-D CNN framework for the spectral–spatial classification of HSI. Furthermore, a 3-D recurrent CNN [24] method was proposed to classify HSI based on 3-D CNN. Xu *et al.* [25] presented the spectral–spatial unified networks with two branches, where one was utilized to extract spectral information with long short-term memory (LSTM) structure and another to learn spatial information with CNN. In [26], a new network combined with CNNs and multiple features learning was introduced to better predict the class labels of HSI pixels. Zhong *et al.* [27] proposed spectral–spatial residual networks (SSRNs), in which residual blocks were used in the process of spectral–spatial feature extraction to improve the classification performance. Shamsolmoali *et al.* [28] built a new framework

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named CNN in network, which exploited spatial information and spectral information and produced high-level features from the original HSIs. The superpixel-based deep learning method also played an important role in HSI classification [29]. Recently, Zhang *et al.* [30] developed a 3-D densely connected convolutional network (3D-DenseNet) for HSI classification, having a deeper structure than 3-D CNN in order to learn more robust spectral–spatial features. Roy *et al.* [31] focused on a new 3-D–2-D CNN feature hierarchy network, referred to as HybridSN, to further learn more abstract-level spatial representation and reduce the complexity of the model. Paoletti *et al.* [32] gave a deep pyramidal residual network to extract multiscale spatial feature for HSI classification. Mu *et al.* [33] built a multiscale and multilevel spectral–spatial feature fusion network for HSI classification, using different scale neighborhood blocks as the input of the network to effectively extract feature. Roy *et al.* [34] designed an end-to-end spectral–spatial squeeze-and-excitation residual bag-of-feature learning framework for HSI classification. Again, more deep learning methods for HSI classification have also emerged in the literature, including CNN with Markov random fields [35], cascaded dual-scale crossover network [36], generative adversarial network [37], and spectral–spatial LSTM networks [38], [39].

Nevertheless, the above-mentioned methods utilize all possible information in the data regardless of whether the information is useful. It is likely that some useless information is considered, which will result in a waste of computing resources. Consequently, we prefer to concern the important details and to suppress some unimportant information of the target regions.

Based on this point, some researchers have introduced the attention mechanism to track the location of interest [40], [41]. This mechanism highlights the sensitivity to features containing the most valuable information. For example, attention inception neural networks (AI-Net) architecture was learned adaptively by dynamic routing between attention initiation modules [42]. Dong *et al.* [43] discussed band attention convolutional networks for HSI classification, which selected suitable parts from hundreds of bands adaptively. Lorenzo *et al.* [44] presented a new attention-based CNN by reusing the convolutional activations at different depths and identifying the most informative regions of the spectral with the gating mechanisms. Fang *et al.* [45] proposed a 3D-DenseNet with spectralwise attention mechanism for HSI classification. Wang *et al.* [46] devised a band attention network for HSI classification, which could undertake both band weighting and band selection tasks. Mou and Zhu [47] introduced an effective end-to-end trainable spectral attention module to improve networks by discriminating the significance of different spectral bands. However, these works only choose the important information of spectral bands and do not reflect the crucial information of spatial location. This may result in the omission of some important spatial information. To this end, Mei *et al.* [48] designed a joint network with a spectral attention bidirectional RNN branch and a spatial attention CNN branch based on PCA to extract spectral–spatial features for HSI classification. In [49], the spectral–spatial attention module was presented, which could learn and select more meaningful feature to improve the accuracy of classification. However, these attention mechanisms are mainly implemented after feature learning, which may result in the information loss of raw input and thereby limit the HSI classification performance.

To overcome the above-mentioned limitations, we develop a novel framework by devising a feature-grouped network with spectral–spatial connected attention mechanism (FG-SSCA). As shown in Fig. 1, the proposed FG-SSCA comprises two crucial modules: the spectral–spatial connected attention (SSCA) module and the feature-grouped network (FGN). Different from the existing works, the SSCA mechanism directly acts on the spectral and spatial positions of the original HSI input. The purpose is to weight the importance of raw input, including reflecting the importance of spectral bands and the better spatial relationship between neighboring pixels. Another different point is that it contains two attention modules (spectral attention and spatial attention) arranged adjacently in a continuous manner, while other methods with spectral and spatial attention are not contiguous. This adjacent strategy helps to highlight useful and valuable information and suppress unnecessary ones from original HSI input. Accordingly, the SSCA module would effectively enhance the information flow by learning which information emphasized within the network and thereby obtain excellent performance. Not only that, we also scheme an FGN based on multiple residual blocks as the spectral–spatial feature extractor. In this step, the coarsely extracted feature maps by the previous SSCA module are segmented into some groups in sequence along the direction of spectral dimension, each of which has the same number of channels. Then, the high-level spectral–spatial features are acquired by this network. This feature-grouped strategy could fully use the redundancy and difference of channels and capture finer features, which further improves generalization performance. Combining these novel ideas, we have implemented an ideal end-to-end FG-SSCA framework and make huge successes in HSI classification. The main contributions are summarized as follows.

1) We present an effective SSCA mechanism by acting on the spectral and spatial positions of the original HSI input continuously. This module could concentrate on different spectral bands and spatial context adaptively by multiplying the importance weights of each channel and each spatial pixel and thereby could strengthen the distinguishing ability of spectral bands and learn the better spatial relationship between neighboring pixels.
2) We design an FGN to further extract spectral–spatial features through the way to parallel residual learning. This structure divides the coarsely extracted feature maps into several groups sequentially according to the direction of spectral dimension, and then, the high-level spectral–spatial features are continuously captured by the spectral and spatial residual modules for each group. This feature-grouped way could fully utilize the redundancy and difference of bands and obtain more available information.

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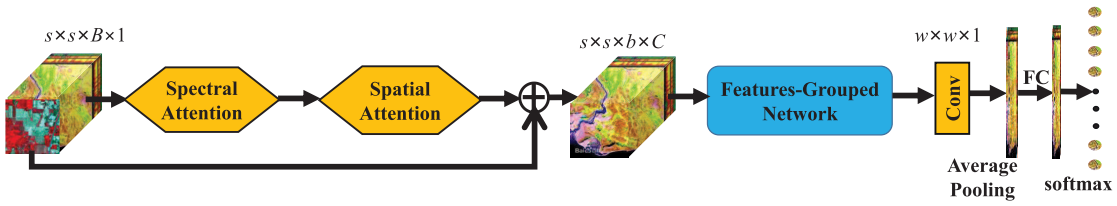


Fig. 1. Overview of the proposed FG-SSCA framework for HSI classification.

3) The integration of the SSCA module and FGN forms the entire FG-SSCA framework, which could effectively promote the HSI classification performance. A series of experiments on several popular HSI data sets is carried out. Experimental results verify the effectiveness and superiority of the proposed method compared with the existing approaches for HSI classification.

The rest of this article is organized as follows. Section II proposed the HSI classification framework and discussed its details. Experiment results and discussions are analyzed about the network in Section III. A summary and future directions of this study are discussed in Section IV.

II. Proposed Methodology

In this section, we explicate the FG-SSCA framework by devising the SSCA module and FGN in detail.

A. Overall FG-SSCA Framework

Given a completed HSI with B spectral bands and $H \times W$ pixels in each band, the original input spatial scale of the network is set as $s \times s$ and the number of spectral bands remains the same as the HSI, i.e., the input is the size of $s \times s \times B$. The entire FG-SSCA framework mainly contains three structures. First, through building the attention modules continuously, we select the important spectral bands from the original input HSI and then mark the spatial locations of interest in these crucial bands. This process is referred to as SSCA module. Subsequently, the coarsely captured features from the SSCA module are embedded into FGN to further extract high-level and fine features. Finally, the classification results can be easily obtained via a fully connected (FC) layer and a softmax layer. The primary FG-SSCA framework is shown in Fig. 1.

In the proposed FG-SSCA method, all the available labeled data are separated into three groups: training group, validation group, and test group for each data set. The training group $\boldsymbol{Z}_1$ and the corresponding label $\boldsymbol{Y}_1$ are used to update the network parameters. The role of the validation group $\boldsymbol{Z}_2$ and the corresponding label $\boldsymbol{Y}_2$ is to monitor the intermediate model generated during the training process and result in an optimal network model. The test group $\boldsymbol{Z}_3$ and the label $\boldsymbol{Y}_3$ are to evaluate the generalization performance of the proposed method. During the whole training process, the cross-entropy loss is adopted, which is defined as

$$\text{Loss} = -\frac{1}{L}\sum_{i=0}^{L}\left(y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right) \quad (1)$$

where L denotes the categories number and y_i and $\hat{y}_i$ represent the ground-truth and predicted labels, respectively.

In the following, we will gradually introduce the principles and implementation steps for each module of the proposed FG-SSCA framework.

B. SSCA Module

The proposed SSCA module comprises two crucial attention modules (spectral attention and spatial attention), which could adaptively focus on different spectral bands and spatial context by multiplying the significant weights of each spectral band and each spatial pixel. In this way, it could enhance the distinguishing ability of spectral bands and learn the superior spatial relationship between the neighboring pixels. As shown in Fig. 2, assume that $\boldsymbol{F} \in \mathbb{R}^{s\times s\times B}$ is the raw HSI input of SSCA module. Then, the overall process could be summarized as follows:

$$\begin{aligned} \boldsymbol{F}_1 &= \boldsymbol{M}_b(\boldsymbol{F}) \otimes \boldsymbol{F} \\ \boldsymbol{F}_2 &= \boldsymbol{M}_s(\boldsymbol{F}_1) \otimes \boldsymbol{F}_1 \end{aligned} \quad (2)$$

where $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$, respectively, stand for the outputs of spectral-band attention and spatial attention, $\boldsymbol{M}_b(\cdot)$ represents the operation of attention extraction in the spectral dimension and the size of output is set as $1 \times 1 \times B$, $\boldsymbol{M}_s(\cdot)$ is the operation of spatial attention extraction and its output is the size of $s \times s \times 1$, and $\otimes$ refers to the elementwise multiplication. Later, we will interpret each attention module in detail.

Specifically, the input of the spectral attention module is aggregated by global average pooling (AvgPool) and global max pooling (MaxPool), respectively, and two 1-D vectors are obtained. These two vectors are added together, and then, two FC layers are followed to produce the spectral band weight $\boldsymbol{M}_b(\boldsymbol{F})$, which could give the response on the importance of each pixel. Mathematically, suppose that $\boldsymbol{F}^b_{\text{avg}}$ and $\boldsymbol{F}^b_{\text{max}}$ represent the average-pooling features and the max-pooling features, respectively. In order to reduce the number of parameter, the sizes of two FC layers are set as $1 \times 1 \times \lfloor B/r \rfloor$ and $1 \times 1 \times B$, where r denotes the reduction ratio and $\lfloor \cdot \rfloor$ represents downward rounding. The whole process is described as

$$\begin{aligned} \boldsymbol{M}_b(\boldsymbol{F}) &= \text{FC}(\text{AvgPool}(\boldsymbol{F}) + \text{MaxPool}(\boldsymbol{F})) \\ &= \delta\left(\boldsymbol{W}_1\boldsymbol{\sigma}\left(\boldsymbol{W}_0\left(\boldsymbol{F}^b_{\text{avg}} + \boldsymbol{F}^b_{\text{max}}\right)\right)\right) \end{aligned} \quad (3)$$

where $\delta(\cdot)$ denotes the sigmoid activation function, $\sigma(\cdot)$ is the rectified linear unit (ReLU) activation, $\boldsymbol{W}_0$ and $\boldsymbol{W}_1$ are the weights of fully connection layers, and FC represents fully connection operation. The final spectral attention feature maps are achieved via multiplying the weight by the raw input,

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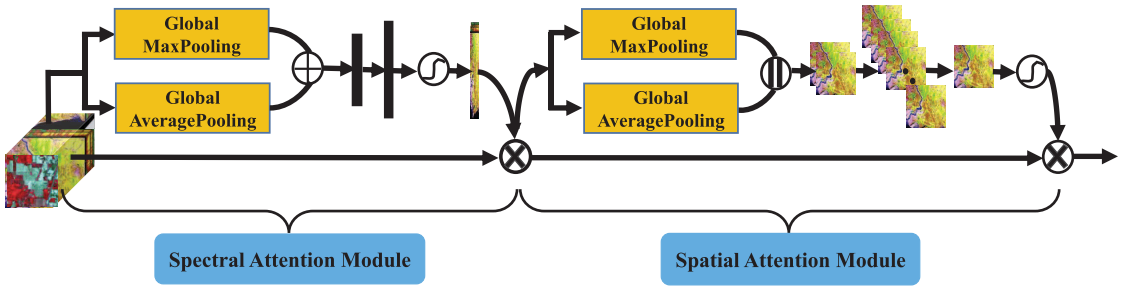


Fig. 2. Flowchart of SSCA module.

recorded as $\boldsymbol{F}_1$ [see (2)]. The specific diagram is shown in the spectral attention module of Fig. 2.

Due to the high correlation of HSI, we design a spatial attention module. Different from spectral attention, the spatial attention module emphasizes the important spatial position of the feature map. To implement this module, global average pooling and global max pooling are utilized to obtain two 2-D feature maps and regard them as two channels, denoted as

$$\begin{aligned}\boldsymbol{F}_c &= \left[\text{AvgPool}(\boldsymbol{F}_1); \text{MaxPool}(\boldsymbol{F}_1)\right] \\ &= \left[\boldsymbol{F}^s_{\text{avg}}; \boldsymbol{F}^s_{\text{max}}\right] \end{aligned} \tag{4}$$

where $\boldsymbol{F}^s_{\text{avg}}, \boldsymbol{F}^s_{\text{max}} \in \mathbb{R}^{s\times s\times 1\times 1}$. Afterward, convolution operations are implemented with C kernels, i.e.,

$$\boldsymbol{F}_{\text{conv}} = \sigma(\boldsymbol{F}_c * \boldsymbol{W}_{a\times a\times 1} + \boldsymbol{b}_{a\times a\times 1}) \tag{5}$$

where "*" denotes the convolution operation, $\boldsymbol{W}_{a\times a\times 1}$ denotes the convolution kernel with the size $a \times a \times 1$, and $\boldsymbol{b}_{a\times a\times 1}$ represents bias. In addition, to ensure that the final feature maps are consistent with the spectral attention feature maps in the spatial dimension, we set "padding = 1" in the process of convolution with 1 kernel. The spatial weight $\boldsymbol{M}_s(\boldsymbol{F}_1)$ is normalized to [0, 1] by a specific sigmoid function. This could be formulated as

$$\boldsymbol{M}_s(\boldsymbol{F}_1) = \delta(\boldsymbol{F}_{\text{conv}} * \boldsymbol{W}_{a\times a\times 1} + \boldsymbol{b}_{a\times a\times 1}). \tag{6}$$

Ultimately, the output features of SSCA are generated by multiplying this weight with the spectral attention feature maps, denoted by $\boldsymbol{F}_2$ [see (2)]. The structure of spatial attention module is shown in Fig. 2.

To sum up, the spatial attention and spectral attention complement each other and promote feature extraction further for HSI classification. With the SSCA module, the representation ability of HSI is heightened by attending important information and suppressing insignificant information of raw input.

C. Feature-Grouped Network

As HSI has dozens to hundreds of bands, we expect to adaptively find the relative low-dimensional representation of HSI. To this end, an FGN is devised to extract sufficient high-level features, which mainly exploits the spectral residual blocks and spatial residual blocks. Fig. 3 shows the computation process of FGN.

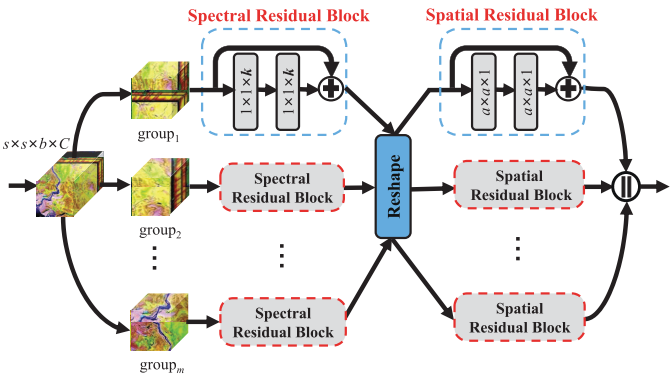


Fig. 3. Illustration of FGN module.

Concretely, we initially use a convolution kernel with the size $1 \times 1 \times h$ and a stride with 2 to implement the dimension reduction written by $\boldsymbol{F}_3$, remaining some essential band information and simultaneously eliminating some nonsignificant band information. This step could obtain the features coarsely. Furthermore, if the difference between feature maps is relatively larger, it is difficult to produce the desired feature representation by convolution operation directly. This would affect the final classification results. For this reason, we divide the coarsely feature maps $\boldsymbol{F}_3$ into m groups in sequence along the direction of spectral dimension, each of which has the same number of channels (here, $m \le b$, where b is the input spectral dimension of FGN and m represents the number of groups).

After that, spectral–spatial features extraction is implemented on the m groups feature maps with m spectral residual blocks and m spatial residual blocks. For each spectral residual block, two convolution kernels with the size $1 \times 1 \times k$ are adopted. Mathematically, this spectral residual block is illustrated as follows:

$$\begin{cases} \boldsymbol{X}^{u+2}_{\text{group}_i} = \boldsymbol{X}^{u}_{\text{group}_i} + f_1\left(\boldsymbol{X}^{u}_{\text{group}_i}; \boldsymbol{\theta}_1\right) \\ f_1\left(\boldsymbol{X}^{u}_{\text{group}_i}; \boldsymbol{\theta}_1\right) = \sigma\left(\boldsymbol{X}^{u+1}_{\text{group}_i}\right) * \boldsymbol{X}^{u+2}_{1\times 1\times k} + \boldsymbol{b}^{u+2}_{1\times 1\times k} \\ \boldsymbol{X}^{u+1}_{\text{group}_i} = \sigma\left(\boldsymbol{X}^{u}_{\text{group}_i}\right) * \boldsymbol{W}^{u+1}_{1\times 1\times k} + \boldsymbol{b}^{u+1}_{1\times 1\times k} \end{cases} \tag{7}$$

where $\boldsymbol{\theta}_1 = \{\boldsymbol{W}^{u+1}_{1\times 1\times k}, \boldsymbol{W}^{u+2}_{1\times 1\times k}, \boldsymbol{b}^{u+1}_{1\times 1\times k}, \boldsymbol{b}^{u+2}_{1\times 1\times k}\}$ is the parameters of the spectral residual module, $\boldsymbol{W}^{u+1}_{1\times 1\times k}$ and $\boldsymbol{b}^{u+1}_{1\times 1\times k}$ represent the spectral convolution kernels and biases in the $(u+1)$th layer, respectively, $f_1(\boldsymbol{X}^{u}_{\text{group}_i}; \boldsymbol{\theta}_1)$ is a spectral residual function, group_i stand for spectral–spatial feature maps of

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the ith group, and $\boldsymbol{X}^{u}_{\text{group}_i}$ and $\boldsymbol{X}^{u+2}_{\text{group}_i}$ refer to the input and output of spectral residual block in the ith group, respectively. The resulting features are used as the input of spatial feature extraction after deformation operation and spatial dimension reduction operation.

Similarly, for each spatial residual block, two convolution kernels with the size $a \times a \times 1$ are used. This spatial residual architecture is formulated as

$$\begin{cases} \boldsymbol{Y}^{r+2}_{\text{group}_i} = \boldsymbol{Y}^{r}_{\text{group}_i} + f_2\left(\boldsymbol{Y}^{r}_{\text{group}_i}; \theta_2\right) \\ f_2\left(\boldsymbol{Y}^{r}_{\text{group}_i}; \boldsymbol{\theta}_2\right) = \sigma\left(\boldsymbol{Y}^{r+1}_{\text{group}_i}\right) * \boldsymbol{W}^{r+2}_{a\times a\times 1} + \boldsymbol{b}^{r+2}_{a\times a\times 1} \\ \boldsymbol{Y}^{r+1}_{\text{group}_i} = \sigma\left(\boldsymbol{Y}^{r}_{\text{group}_i}\right) * \boldsymbol{W}^{r+1}_{a\times a\times 1} + \boldsymbol{b}^{r+1}_{a\times a\times 1} \end{cases} \tag{8}$$

where $\boldsymbol{\theta}_2 = \{\boldsymbol{W}^{r+1}_{a\times a\times 1}, \boldsymbol{W}^{r+2}_{a\times a\times 1}, \boldsymbol{b}^{r+1}_{a\times a\times 1}, \boldsymbol{b}^{r+2}_{a\times a\times 1}\}$ is the parameters in the spatial residual module, $\boldsymbol{W}^{r+1}_{a\times a\times 1}$ and $\boldsymbol{b}^{r+1}_{a\times a\times 1}$ represent the spatial convolution kernels and biases in the $(r+1)$th layer, respectively, $f_2(\boldsymbol{Y}^{r}_{\text{group}_i}; \boldsymbol{\theta}_2)$ denotes a spatial residual function, and $\boldsymbol{Y}^{r}_{\text{group}_i}$ and $\boldsymbol{Y}^{r+2}_{\text{group}_i}$ denote the input and output of spatial residual block in the ith group, respectively. At the end of FGN, the spectral–spatial features of m groups are merged through a simple concatenated operation.

With the SSCA module and FGN, it is easy to effectively capture more finer and high-level HSI features. After these steps, a convolution operator and an average-pooling layer transform the extracted spectral–spatial feature volume with the size of $s^* \times s^* \times C$ to a feature vector with $1 \times 1 \times C$. Then, an FC layer and a softmax function generate an prediction label vector (i.e., $\hat{\boldsymbol{y}} = [\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_L]$), which combines with the ground-truth label $\boldsymbol{y} = \{y_1, y_2, \ldots, y_L\}$ to compute the classification accuracies.

III. Experiments and Analysis

In this section, a series of experiments on some HSI data sets is undertaken to validate the effectiveness of the proposed FG-SSCA framework. All the experiments are implemented on the same configuration, which is the software Keras of python 3.5.2 running on a NVIDIA Quadro K2200(GPU) on Windows 10 based on Intel Xeon CPU E5-1620 v3 with 32-GB main memory.

A. Experimental Setting

1) Data Sets: In the experiments, three commonly used HSI data sets,[1] including Indian Pines (IN) data set, Kennedy Space Center (KSC) data set, and University of Pavia (UP) data set, are utilized to evaluate the performance of the proposed network. The IN data set was collected by the NASA Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) instrument in Northwest Indiana in 1992 and consisted of 145×145 pixels with 200 spectral bands after removing water absorption bands. The ground truth has 16 types of vegetation. The KSC data set was acquired by the AVIRIS from Florida in 1996. The original image cube is the size of $512 \times 614 \times 176$ after removing water absorption bands and low signal-to-noise ratio (SNR) bands. The ground truth composed of 13 categories. The UP data set, possessing 103 spectral bands with 610×340 pixels in each band, was gathered by the Reflective Optical System Imaging Spectrometer (ROSIS) from Northern Italy in 2001 and contains nine classes urban land covers. The detailed information of these data sets is listed in Table I.

2) Parameters Setting: In the experiments, the batch size is assigned to 16 and the size of the convolution kernel is set as $C = 32$. The input spatial size $s \times s$ is 9×9. The number of groups is set as $m = 3$ for the IN and KSC data sets and $m = 2$ for the UP data set. Select the reduction ratio $r = 4$ for the IN and KSC data sets and $r = 3$ for the UP data set empirically. The influence of these hyparameters will be analyzed detailedly in Section III-C. In addition, we adopt the "Adam" optimizer [50] in the training process. The initial learning rate is given with 10^{-3} and linearly decays for every 40 iterations. Furthermore, the number of maximal training epochs is set at 180.

3) Evaluation Indices: Three common indices [51], the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa), are adopted to perform the quantitative evaluation in the HSI classification. To this end, we construct the confusion matrix through the real category information in the data and the predicted category information in the proposed method. Suppose that HSI has a total of N samples and n categories, and then, $\boldsymbol{A} = (a_{ij})_{n\times n}$ is the confusion matrix, where the element a_{ij} represents the number of samples in categories j that are classified into categories i. Therefore, OA is defined as the ratio of the number of samples that are accurately classified and the total number of samples, i.e.,

$$\text{OA} = \frac{\sum_{i=1}^{n} a_{ii}}{N} \times 100\% \tag{9}$$

where a_{ii} represents the diagonal elements of the corresponding confusion matrix. Also, AA stands for the average value on the percentage of pixels accurately classified in each category in the total number of categories. Mathematically, this is defined as

$$\text{AA} = \sum_{i=1}^{n} \frac{a_{ii}}{n \sum_{j=1}^{n} a_{ij}} \times 100\% \tag{10}$$

where Kappa coefficient is another index to measure the classification performance. The mathematical expression is described as

$$\text{Kappa} = \frac{N \sum_{i=1}^{n} a_{ii} - \sum_{i=1}^{n} (a_{i+} a_{+i})}{N^2 - \sum_{i=1}^{n} (a_{i+} a_{+i})} \tag{11}$$

where a_{i+} and a_{+i} refer to the sum of the ith row and column elements in the confusion matrix, respectively. Generally, larger OA, AA, and Kappa reflect better classification results.

B. Comparison Results

In this section, we compare the proposed FG-SSCA network with some popular classification methods, including 1-D CNN [20], AI-Net [42], 2-D CNN [21], 3-D CNN [23], SSRN [27], 3D-DenseNet [30], HybridSN [31], and SSAN [48]. In the experiments, 20%, 10%, and 70% of the labeled

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

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TABLE I
INFORMATION OF HSI DATA SETS USED FOR EXPERIMENTS

IN			KSC			UP		
No.	Class	Number of sample	No.	Class	Number of sample	No.	Class	Number of sample
1	Alfalfa	46	1	Scrub	347	1	Asphalt	6631
2	Corn-notill	1428	2	Willow swamp	243	2	Meadows	18649
3	Corn-mintill	830	3	CP hammock	256	3	Gravel	2099
4	Corn	237	4	Slash pine	252	4	Trees	3064
5	Grass-pasture	483	5	Oak/broadleaf	161	5	Painted metal sheets	1345
6	Grass-tree	730	6	Hardwood	229	6	Bare soil	5029
7	Grass-pasture-mowed	28	7	Swamp	105	7	Bitumen	1330
8	Hay-windrowed	478	8	Graminoid marsh	390	8	Self-blocking bricks	3682
9	Oats	20	9	Spartina marsh	520	9	Shadows	976
10	Soybean-notill	972	10	Cattail marsh	404	/	/	/
11	Soybean-mintill	2455	11	Salt marsh	419	/	/	/
12	Soybean-clean	593	12	Muds flats	503	/	/	/
13	Wheat	205	13	Water	927	/	/	/
14	Woods	1265	/	/	/	/	/	/
15	Buildings-Grass-Trees	386	/	/	/	/	/	/
16	Stone-Steel-Towers	93	/	/	/	/	/	/
Total	/	10249	Total	/	5211	Total	/	42776

TABLE II
CLASSIFICATION RESULTS WITH DIFFERENT METHODS FOR IN

Class	1D-CNN	AI-Net	2D-CNN	3D-CNN	SSRN	3D-DenseNet	HybridSN	SSAN	FG-SSCA
1	66.66	96.97	88.88	**100**	97.82	**100**	**100**	**100**	**100**
2	91.68	97.06	99.29	98.90	99.17	97.88	97.73	99.04	**99.71**
3	75.00	99.12	99.27	99.65	99.53	99.64	99.55	98.09	**100**
4	67.04	95.65	97.15	98.75	97.79	99.38	**100**	99.38	99.40
5	86.09	**100**	97.92	98.56	99.24	99.70	**100**	99.04	**100**
6	97.43	98.42	**100**	99.80	99.51	**100**	99.83	98.26	**100**
7	81.81	**100**	**100**	94.73	98.70	81.82	**100**	80.00	**100**
8	99.47	**100**	**100**	**100**	99.85	99.70	**100**	**100**	**100**
9	56.25	**100**	68.75	**100**	98.50	93.75	**100**	**100**	**100**
10	87.41	96.01	97.29	99.55	98.74	**99.71**	99.61	**100**	99.70
11	86.22	99.40	98.19	98.83	99.30	99.42	**99.85**	99.25	99.83
12	87.50	88.45	98.21	96.52	98.43	99.25	98.52	99.28	**99.76**
13	96.34	**100**	**100**	**100**	**100**	**100**	**100**	97.93	**100**
14	98.81	98.66	**100**	99.77	99.31	99.55	**100**	**100**	99.77
15	55.18	97.12	96.22	**100**	99.20	98.87	**100**	**100**	98.20
16	89.18	97.05	98.64	98.57	97.82	95.38	94.59	**100**	**100**
OA(%)	88.34 ± 0.56	96.97 ± 0.56	98.04 ± 0.50	99.12 ± 0.11	99.19 ± 0.26	99.20 ± 0.20	99.33 ± 0.10	99.20 ± 0.12	$\mathbf{99.79 \pm 0.02}$
AA(%)	82.63 ± 0.61	97.58 ± 0.92	97.86 ± 0.37	99.08 ± 0.09	98.93 ± 0.59	97.75 ± 0.58	99.30 ± 0.11	97.88 ± 0.58	$\mathbf{99.77 \pm 0.02}$
Kappa(%)	86.63 ± 0.83	96.54 ± 0.56	97.77 ± 0.59	99.07 ± 0.24	99.07 ± 0.30	99.09 ± 0.22	99.24 ± 0.12	99.07 ± 0.13	$\mathbf{99.76 \pm 0.02}$

TABLE III
CLASSIFICATION RESULTS WITH DIFFERENT METHODS FOR KSC

Class	1D-CNN	AI-Net	2D-CNN	3D-CNN	SSRN	3D-DenseNet	HybridSN	SSAN	FG-SSCA
1	98.31	99.43	99.43	**100**	99.24	**100**	99.81	94.44	**100**
2	86.08	**100**	90.20	**100**	99.40	**100**	**100**	92.15	**100**
3	83.33	95.65	93.13	96.35	**100**	**100**	97.21	96.57	99.43
4	78.10	89.66	83.58	97.64	98.32	**100**	98.86	87.44	**100**
5	54.68	93.94	85.93	93.96	**100**	**100**	**100**	89.31	**100**
6	33.33	87.43	97.81	99.37	**100**	98.83	99.38	82.87	**100**
7	96.42	94.03	96.42	**100**	**100**	**100**	97.30	90.80	**100**
8	90.60	99.34	97.27	**100**	**100**	99.33	**100**	90.83	**100**
9	89.42	**100**	98.31	**100**	**100**	**100**	**100**	97.01	**100**
10	96.28	98.95	95.66	**100**	**100**	**100**	**100**	93.54	**100**
11	98.80	**100**	**100**	**100**	**100**	**100**	**100**	97.98	**100**
12	90.79	98.91	98.75	**100**	**100**	**100**	**100**	97.54	**100**
13	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
OA(%)	89.56 ± 0.76	98.22 ± 0.57	98.56 ± 0.70	99.47 ± 0.49	99.61 ± 0.22	99.75 ± 0.06	99.53 ± 0.09	94.74 ± 0.64	$\mathbf{99.96 \pm 0.02}$
AA(%)	84.32 ± 1.04	96.82 ± 0.66	97.55 ± 0.86	99.02 ± 0.66	99.33 ± 0.57	99.63 ± 0.11	99.33 ± 0.10	93.06 ± 0.23	$\mathbf{99.94 \pm 0.03}$
Kappa(%)	88.37 ± 0.99	98.02 ± 0.63	98.39 ± 0.78	99.42 ± 0.44	99.56 ± 0.25	99.73 ± 0.07	99.48 ± 0.10	94.24 ± 0.44	$\mathbf{99.96 \pm 0.02}$

HSI data are randomly assigned to the training set, validation set, and test set for the IN and KSC data sets, respectively. Analogously, the ratio is 10%, 10%, and 80% for the UP data set. More cases of other different proportions on the training samples will be discussed in detail in Section III-C. We randomly select training samples with ten trials for IN, KSC, and UP data sets and take the average values as the final classification accuracies.

Tables II–IV show the quantization comparison of accuracies with different methods under three benchmark HSI data sets, where the test results are given in the form of "mean ± variance." According to the values of OAs,

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TABLE IV
CLASSIFICATION RESULTS WITH DIFFERENT METHODS FOR UP

Class	1D-CNN	AI-Net	2D-CNN	3D-CNN	SSRN	3D-DenseNet	HybridSN	SSAN	FG-SSCA
1	94.63	96.35	98.63	99.43	99.92	99.28	**100**	96.07	99.98
2	98.59	99.54	99.93	99.87	99.96	**100**	**100**	99.73	99.94
3	76.39	99.29	90.88	99.07	98.46	97.46	99.31	96.98	**100**
4	96.64	99.14	98.97	99.54	99.69	**100**	99.53	99.88	99.96
5	99.84	99.90	**100**	**100**	99.99	99.91	99.92	**100**	**100**
6	87.90	99.70	99.72	99.47	**99.94**	99.93	**100**	88.12	**100**
7	89.54	96.46	99.46	99.62	99.82	**99.90**	99.58	**100**	**100**
8	92.99	92.83	98.77	98.51	99.22	**99.49**	99.25	99.06	99.93
9	99.55	**100**	98.77	**100**	99.95	**100**	94.72	98.68	**100**
OA(%)	94.68 ± 0.53	98.47 ± 0.59	99.43 ± 0.10	99.58 ± 0.10	99.79 ± 0.09	99.70 ± 0.04	99.75 ± 0.05	98.88 ± 0.17	$\mathbf{99.96 \pm 0.01}$
AA(%)	92.90 ± 0.34	98.23 ± 0.59	99.26 ± 0.12	99.50 ± 0.09	99.66 ± 0.17	99.55 ± 0.04	99.27 ± 0.17	98.91 ± 0.15	$\mathbf{99.97 \pm 0.01}$
Kappa(%)	93.05 ± 0.49	98.01 ± 0.39	99.25 ± 0.13	99.44 ± 0.13	99.72 ± 0.12	99.60 ± 0.03	99.67 ± 0.06	98.51 ± 0.23	$\mathbf{99.94 \pm 0.01}$

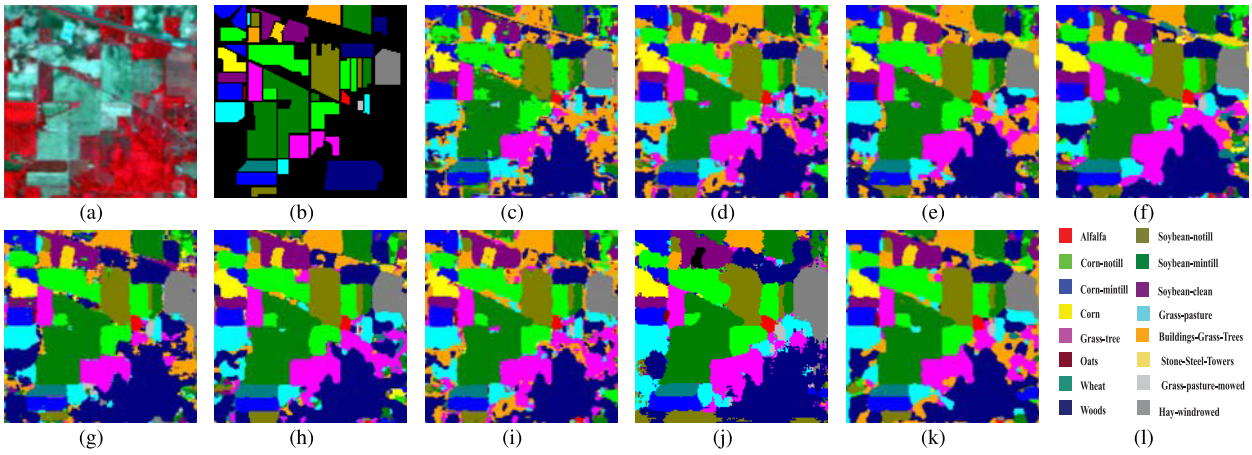


Fig. 4. Visual classification results with different methods for IN. (a) False color. (b) Truth labels. (c) 1-D CNN (88.34%). (d) AI-NET (96.97%). (e) 2-D CNN (98.04%). (f) 3-D CNN (99.12%). (g) 3D-DenseNet (99.20%). (h) SSRN (99.19%). (i) HybridSN (99.33%). (j) SSAN (99.20%). (k) FG-SSCA (99.79%). (l) Colors ground. Here, the values in the brackets are the OAs.

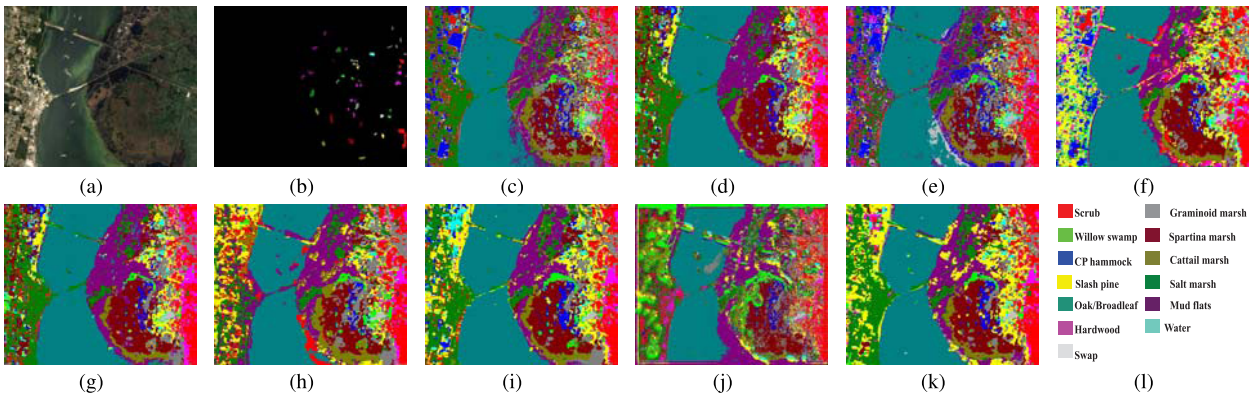


Fig. 5. Visual classification results with different methods for KSC. (a) False color. (b) Truth labels. (c) 1-D CNN (89.56%). (d) AI-NET (98.22%). (e) 2-D CNN (98.56%). (f) 3-D CNN (99.47%). (g) 3D-DenseNet (99.75%). (h) SSRN (99.61%). (i) HybridSN (99.53%). (j) SSAN (94.74%). (k) FG-SSCA (99.96%). (l) Colors ground. Here, the values in the brackets are the OAs.

AAs, and Kappas, it can be seen that the results of the proposed method are remarkably superior to those of 1-D CNN, AI-Net, 2-D CNN, 3-D CNN, SSRN, 3D-DenseNet, HybridSN, and SSAN for these data sets, reflecting the better classification performance of the proposed architecture. One possible reason is that the FG-SSCA network takes advantage of attention module effectively, selecting the important bands and marking the spatial location of interest in which we could know that "where" and "what" need to be emphatically learned in original HSI. Another point is that contextual information of high level could be fully extracted by FGN.

Furthermore, Figs. 4–6 show the corresponding visual images for each of the methods of the best-trained models in these benchmark data sets, including the false-color images, the corresponding ground-truth maps, and visualization images under diverse comparison methods. Apparently, it can be easily observed that the visual images generated by 1-D CNN are

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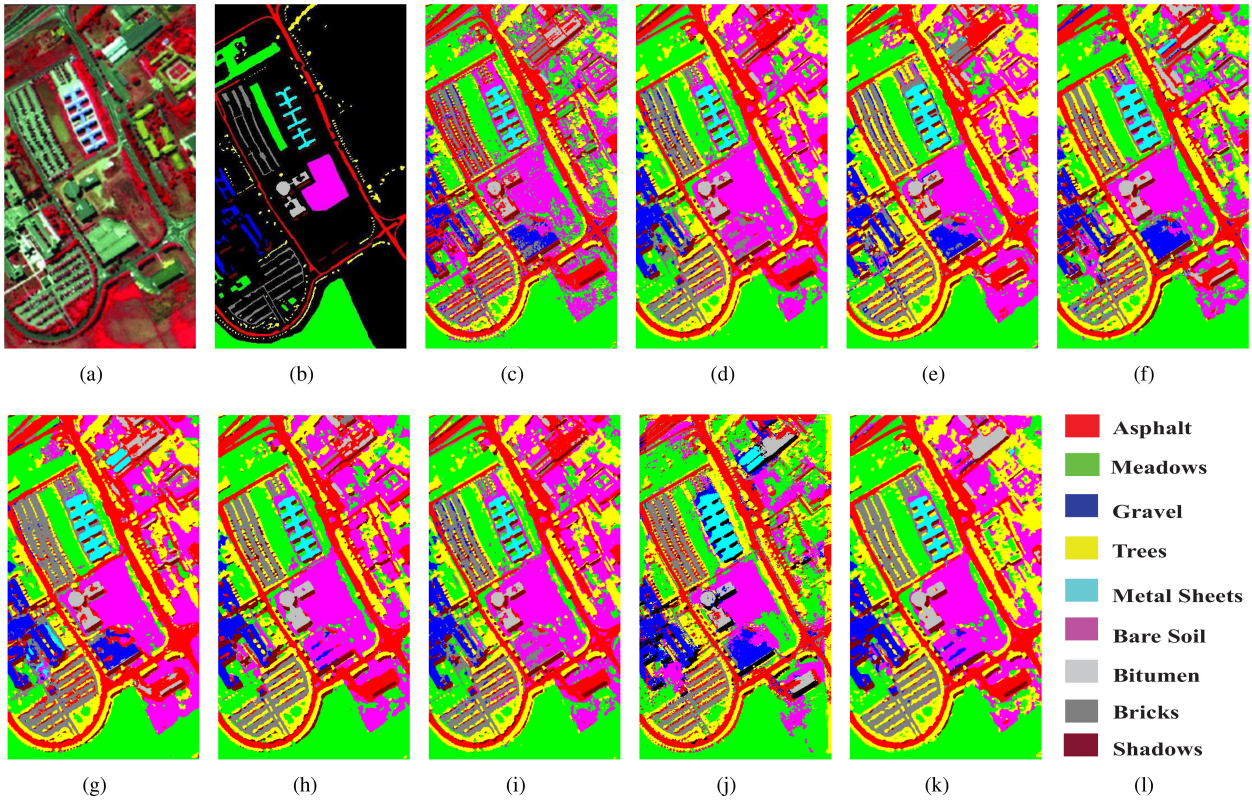


Fig. 6. Visual classification results with different methods for UP. (a) False color. (b) Truth labels. (c) 1-D CNN (94.68%). (d) AI-Net (98.47%). (e) 2-D CNN (99.43%). (f) 3-D CNN (99.58%). (g) 3D-DenseNet (99.70%). (h) SSRN (99.79%). (i) HybridSN (99.75%). (j) SSAN (98.88%). (k) FG-SSCA (99.96%). (l) Colors ground. Here, the values in the brackets are the OAs.

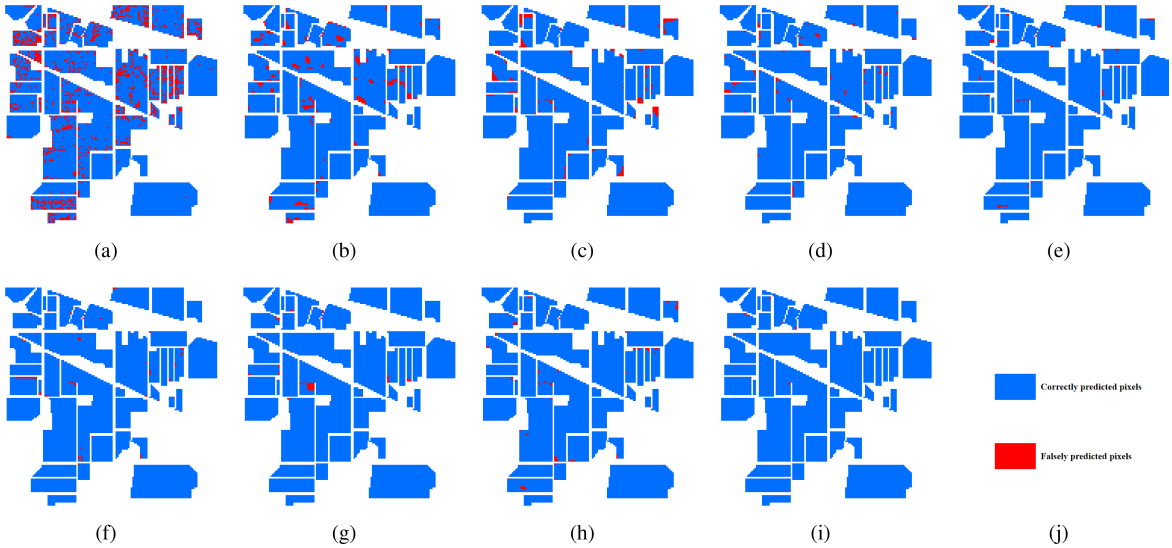


Fig. 7. Error maps with different methods for IN. (a) 1-D CNN (1221). (b) AI-NET (307). (c) 2-D CNN (201). (d) 3-D CNN (92). (e) 3D-DenseNet (78). (f) SSRN (82). (g) HybridSN (61). (h) SSAN (81). (i) FG-SSCA (24). (j) Color labels. The values in the brackets are the number of the misclassification (red color) pixels.

greatly coarse, which have lower classification accuracies and obvious noises. The probable explanation is that the conventional methods could not effectively distill the spectral–spatial features, making the classification effect unsatisfactory. Again, the visual images obtained by AI-Net, 2-D CNN, 3-D CNN, 3D-DenseNet, SSRN, HybridSN, and SSAN under the end-to-end deep learning framework are relatively smooth and exist less noises. In contrast, the proposed method could produce better visualization effects for the three HSIs data sets. One important reason is that the proposed SSCA module could be able to determine more accurately which spectral band is important and what is the interesting part of the spatial position, and the FGN could extract more abstract features information and enhance the classification ability of

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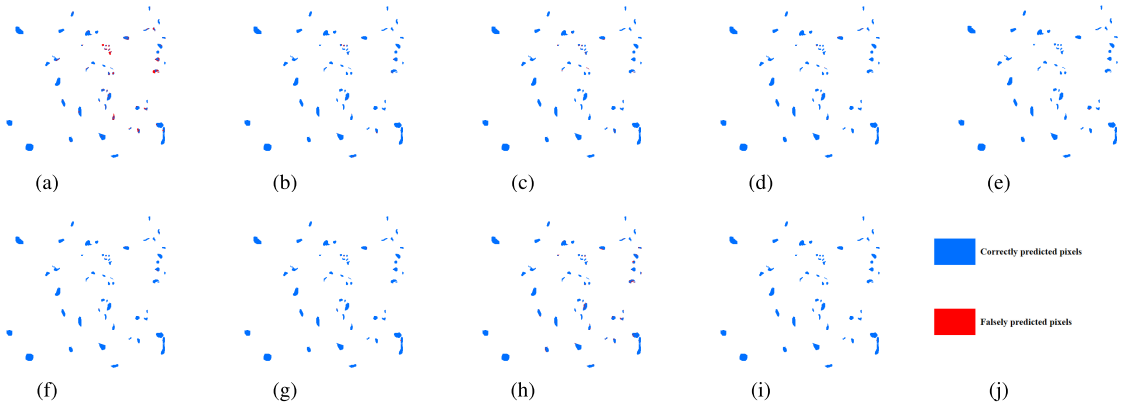


Fig. 8. Error maps with different methods for KSC. (a) 1-D CNN (631). (b) AI-NET (104). (c) 2-D CNN (72). (d) 3-D CNN (28). (e) 3D-DenseNet (12). (f) SSRN (20). (g) HybridSN (25). (h) SSAN (278). (i) FG-SSCA (4). (j) Color labels. The values in the brackets are the number of the misclassification (red color) pixels.

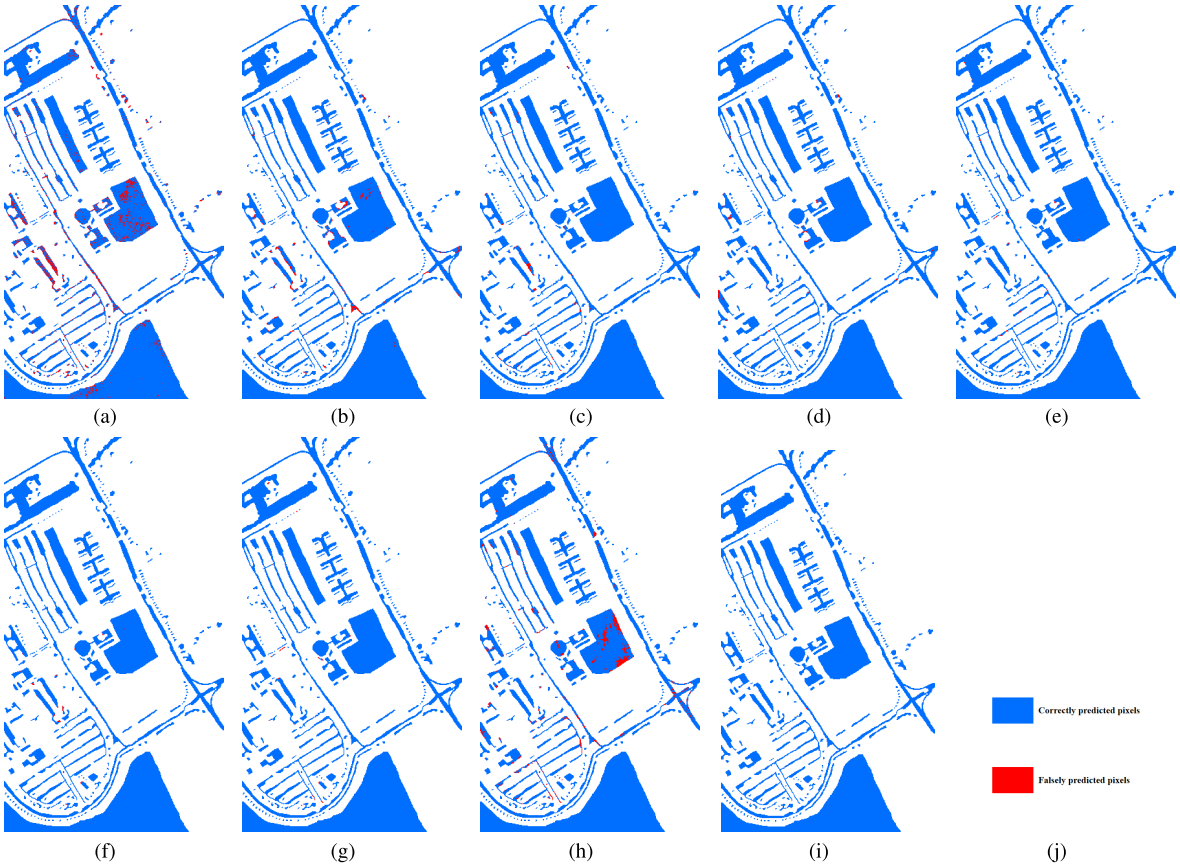


Fig. 9. Error maps with different methods for UP. (a) 1-D CNN (2310). (b) AI-NET (654). (c) 2-D CNN (243). (d) 3-D CNN (201). (e) 3D-DenseNet (124). (f) SSRN (90). (g) HybridSN (103). (h) SSAN (479). (i) FG-SSCA (20). (j) Color labels. The values in the brackets are the number of the misclassification (red color) pixels.

the FG-SSCA network. Moreover, we also plot the error maps by comparing the predicted label of the pixel with the ground-truth label of the same pixel one by one according to Figs. 4–6, except for the black regions (unknown ground-truth labels). The results are shown in Figs. 7–9, where the blue color represents that the predicted label is consistent with the ground-truth label for the same pixel, whereas the red color indicates that the predicted label is not matched with the ground-truth label. The more blue pixels or the less red pixels illustrate the better classification performance. Obviously, the number of red pixels on the proposed method is far less than other methods. This can be further reflected in the number of the misclassification pixels (see in the titles of the figures). Accordingly, these analyses prove that the FG-SSCA method has superior classification performance.

To further illustrate the advantages of the proposed method, we do experiments by reducing the training percentage from 20% to 10% for the IN and KSC data sets and from 10%

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TABLE V
CLASSIFICATION RESULTS BY DIFFERENT METHODS WITH 10% TRAINING SAMPLES FOR IN AND KSC

		1D-CNN	AI-Net	2D-CNN	3D-CNN	SSRN	3D-DenseNet	HybridSN	SSAN	FG-SSCA
IN	OA(%)	75.44 ± 0.51	88.12 ± 0.22	95.97 ± 1.32	97.08 ± 0.74	97.28 ± 0.25	82.66 ± 0.79	98.03 ± 0.94	97.40 ± 0.35	$\mathbf{99.21 \pm 0.02}$
	AA(%)	80.98 ± 1.09	70.23 ± 0.81	96.30 ± 1.03	96.97 ± 0.85	96.88 ± 0.43	82.52 ± 1.16	96.78 ± 0.99	94.42 ± 0.43	$\mathbf{98.88 \pm 0.06}$
	Kappa(%)	71.69 ± 0.55	86.44 ± 0.26	96.00 ± 1.23	96.67 ± 0.84	96.87 ± 0.28	80.23 ± 0.95	97.75 ± 1.08	97.00 ± 0.40	$\mathbf{99.09 \pm 0.02}$
KSC	OA(%)	86.23 ± 0.56	97.20 ± 0.24	97.08 ± 0.55	96.38 ± 0.27	98.87 ± 0.29	91.81 ± 0.37	97.74 ± 0.38	88.82 ± 0.24	$\mathbf{99.42 \pm 0.17}$
	AA(%)	81.78 ± 1.06	95.35 ± 0.29	95.06 ± 0.76	94.83 ± 0.42	98.48 ± 0.37	89.83 ± 0.40	96.15 ± 0.50	84.43 ± 0.41	$\mathbf{99.02 \pm 0.31}$
	Kappa(%)	83.85 ± 1.43	96.88 ± 0.27	96.76 ± 0.61	95.98 ± 0.30	98.77 ± 0.34	90.88 ± 0.42	97.48 ± 0.48	87.55 ± 0.17	$\mathbf{99.35 \pm 0.16}$

TABLE VI
CLASSIFICATION RESULTS BY DIFFERENT METHODS WITH 5% TRAINING SAMPLES FOR UP

		1D-CNN	AI-Net	2D-CNN	3D-CNN	SSRN	3D-DenseNet	HybridSN	SSAN	FG-SSCA
UP	OA(%)	89.45 ± 0.07	97.35 ± 0.09	98.88 ± 0.76	99.04 ± 0.81	99.30 ± 0.05	97.11 ± 0.38	99.32 ± 0.47	96.35 ± 0.12	$\mathbf{99.80 \pm 0.03}$
	AA(%)	86.71 ± 0.11	97.54 ± 0.09	98.78 ± 0.63	98.65 ± 0.98	99.23 ± 0.06	95.43 ± 0.69	98.50 ± 0.96	95.53 ± 0.18	$\mathbf{99.70 \pm 0.03}$
	Kappa(%)	86.13 ± 0.11	96.09 ± 0.12	98.53 ± 1.02	98.72 ± 1.06	99.08 ± 0.11	96.19 ± 0.42	$99,10 \pm 0.63$	95.19 ± 0.15	$\mathbf{99.75 \pm 0.04}$

TABLE VII
TEST ACCURACIES OF DIFFERENT ATTENTION MODULES

		FGN	RCAB+FGN	CSAR+FGN	RAM+FGN	FG-SSCA
OA(%)	IN	99.52 ± 0.06	99.70 ± 0.01	99.48 ± 0.03	99.73 ± 0.03	99.79 ± 0.02
	KSC	99.91 ± 0.02	99.83 ± 0.05	99.89 ± 0.02	99.84 ± 0.06	99.96 ± 0.02
	UP	99.90 ± 0.01	99.88 ± 0.02	99.89 ± 0.02	99.93 ± 0.02	99.96 ± 0.01
AA(%)	IN	99.47 ± 0.09	99.78 ± 0.01	99.72 ± 0.03	99.72 ± 0.02	99.77 ± 0.02
	KSC	99.87 ± 0.03	99.70 ± 0.12	99.86 ± 0.03	99.79 ± 0.07	99.94 ± 0.03
	UP	99.91 ± 0.01	99.82 ± 0.04	99.87 ± 0.02	99.94 ± 0.01	99.94 ± 0.01
Kappa(%)	IN	99.45 ± 0.06	99.66 ± 0.02	99.41 ± 0.11	99.70 ± 0.03	99.76 ± 0.02
	KSC	99.91 ± 0.01	99.82 ± 0.09	99.88 ± 0.04	99.82 ± 0.05	99.96 ± 0.02
	UP	99.87 ± 0.01	99.85 ± 0.02	99.86 ± 0.01	99.92 ± 0.02	99.93 ± 0.01

TABLE VIII
TEST ACCURACIES FOR EACH DATA SET UNDER DIFFERENT SPATIAL SIZES

Sizes		3×3	5×5	7×7	9×9	11×11
IN	OA(%)	93.54 ± 0.18	98.70 ± 0.06	99.64 ± 0.02	$\mathbf{99.79 \pm 0.02}$	99.72 ± 0.05
	AA(%)	94.53 ± 0.05	98.74 ± 0.04	99.69 ± 0.03	99.77 ± 0.02	$\mathbf{99.84 \pm 0.03}$
	Kappa(%)	92.40 ± 0.10	98.52 ± 0.18	99.58 ± 0.03	$\mathbf{99.76 \pm 0.02}$	99.68 ± 0.05
KSC	OA(%)	97.25 ± 0.27	99.34 ± 0.11	99.73 ± 0.09	$\mathbf{99.96 \pm 0.02}$	99.91 ± 0.05
	AA(%)	95.68 ± 0.38	99.20 ± 0.17	99.73 ± 0.15	$\mathbf{99.94 \pm 0.03}$	99.89 ± 0.11
	Kappa(%)	96.94 ± 0.30	99.27 ± 0.13	99.70 ± 0.13	$\mathbf{99.96 \pm 0.02}$	99.91 ± 0.06
UP	OA(%)	97.36 ± 0.11	99.30 ± 0.06	99.84 ± 0.03	$\mathbf{99.96 \pm 0.01}$	99.90 ± 0.03
	AA(%)	97.17 ± 0.08	99.14 ± 0.08	99.82 ± 0.04	$\mathbf{99.97 \pm 0.01}$	99.84 ± 0.06
	Kappa(%)	96.50 ± 0.13	99.07 ± 0.04	99.79 ± 0.02	$\mathbf{99.94 \pm 0.01}$	99.86 ± 0.03

to 5% for the UP data set. The corresponding results for different methods are listed in Tables V and VI. It can be easily observed that the OAs, AAs, and Kappas of the FG-SSCA method are superior to those of the other eight methods remarkably, which indicates the effectiveness of the proposed approach.

In short, the FG-SSCA method can satisfactorily obtain excellent classification performance for HSI.

C. Discussion

In this section, we discuss and analyze some main factors that are likely to influence the classification performance in the proposed framework concretely. The above-mentioned three data sets are still used to explore the variation trend of the classification accuracy.

1) Impact With Different Attention Modules: To verify the effectiveness of the SSCA module, we consider other three attention modules as a comparison, such as residual channel attention module (RCAB) [52], channelwise and spatial attention residual (CSAR) [53], and residual attention module (RAM) [54] (see Fig. 10). In addition, the network without attention module (i.e., only FGN) is also utilized for comparison. Experimental results with various attention methods are shown in Table VII. It can be concluded that the accuracies of adopting the proposed method outperform those of other methods. Simultaneously, it also indicates that the use of SSCA is beneficial to the classification of HSI. The possible interpretation is that the RAM module marks only the important spectral band information and the spatial features are not fully utilized when the features are extracted. For the RCAB and CSAR modules, features concatenation or addition may change the distribution of features and hinder information extraction after using the spectral attention block and the spatial attention block, respectively. However, the SSCA module we designed could mark important bands and spatial locations better in comparison with other attention modules.

2) Effect of Various Numbers of Groups in the FGN: To sufficiently extract features, the resulting coarsely features from SSCA are divided into m groups along with the spectral direction and take them as the input of FGN. To consider the influence of feature-grouped, we empirically conduct the experiments with $m = 1, 3, 5, 7, 9$ groups for IN and KSC data sets and $m = 1, 2, 3, 4, 5$ for UP data set. The corresponding results are given in Fig. 11. Evidently, the test accuracies reach the highest when the number of groups is $m = 3$ for the IN and KSC data sets and $m = 2$ for the UP data set. Consequently, it can get that the "grouped" strategy can capture features more effectively, which is useful to the classification of HSI. In the experiments, we empirically select the number of groups m with 3 for the former two data sets and 2 for the last data set.

3) Influence on the Input Spatial Size: In order to select a better input spatial size $s \times s$, we have carried out experiments on the spatial sizes of 3×3, 5×5, 7×7, 9×9, and 11×11. The final experimental results are reported in Table VIII. It is easy to notice that the test accuracies increase at first and tend to stable as the increase of spatial size. When the input size is 9×9, the accuracies are the best, and therefore, we choose

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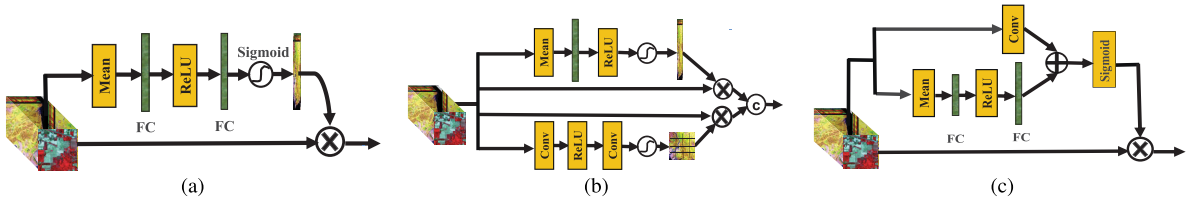


Fig. 10. Comparison of different attention methods. (a) RCAB. (b) CSAR. (c) RAM.

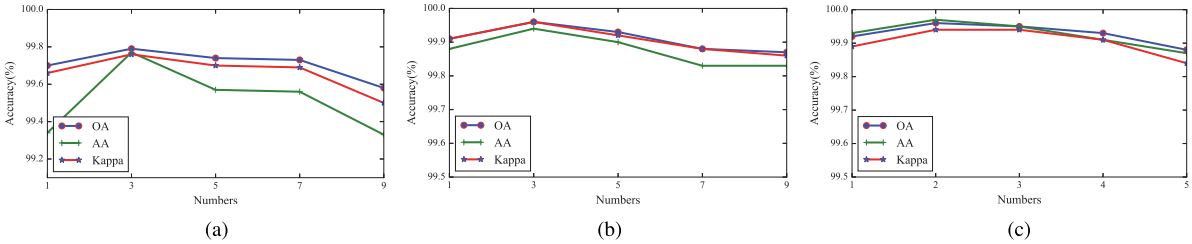


Fig. 11. Test accuracies of different features grouped for three data sets. (a) IN. (b) KSC. (c) UP.

TABLE IX
TEST ACCURACIES UNDER DIFFERENT TRAINING SAMPLES FOR IN AND KSC

training samples		5%	10%	15%	20%	25%
IN	OA(%)	97.45 ± 0.06	99.21 ± 0.02	99.67 ± 0.02	99.79 ± 0.02	99.86 ± 0.07
	AA(%)	97.43 ± 0.15	98.88 ± 0.06	99.66 ± 0.05	99.77 ± 0.02	99.72 ± 0.15
	Kappa(%)	97.11 ± 0.07	99.09 ± 0.02	99.63 ± 0.03	99.76 ± 0.02	99.85 ± 0.07
KSC	OA(%)	98.78 ± 0.16	99.42 ± 0.17	99.68 ± 0.04	99.96 ± 0.02	99.96 ± 0.02
	AA(%)	98.14 ± 0.26	99.02 ± 0.31	99.46 ± 0.45	99.94 ± 0.03	99.95 ± 0.04
	Kappa(%)	98.64 ± 0.18	99.35 ± 0.16	99.65 ± 0.05	99.96 ± 0.02	99.97 ± 0.04

TABLE X
TEST ACCURACIES UNDER DIFFERENT TRAINING SAMPLES FOR UP

training samples		1%	2.5%	5%	7.5%	10%	12.5%
UP	OA(%)	98.74 ± 0.20	99.29 ± 0.06	99.80 ± 0.03	99.85 ± 0.03	99.96 ± 0.01	99.98 ± 0.00
	AA(%)	98.14 ± 0.07	99.16 ± 0.07	99.75 ± 0.03	99.78 ± 0.03	99.97 ± 0.01	99.99 ± 0.00
	Kappa(%)	98.33 ± 0.20	99.05 ± 0.07	99.70 ± 0.04	99.81 ± 0.04	99.94 ± 0.01	99.98 ± 0.01

TABLE XI
IMPACT OF DIFFERENT VALUES r FOR THREE DATA SETS

		$r=1$	$r=2$	$r=3$	$r=4$	$r=5$	$r=6$	$r=7$	$r=8$
IN	OA(%)	99.10 ± 0.17	99.19 ± 0.12	98.81 ± 0.09	$\mathbf{99.21 \pm 0.02}$	99.00 ± 0.17	98.91 ± 0.22	98.71 ± 0.14	98.71 ± 0.10
KSC	OA(%)	99.22 ± 0.15	99.36 ± 0.16	99.22 ± 0.15	$\mathbf{99.42 \pm 0.17}$	99.32 ± 0.17	99.21 ± 0.196	99.15 ± 0.04	99.13 ± 0.09
UP	OA(%)	99.62 ± 0.04	99.68 ± 0.04	$\mathbf{99.80 \pm 0.03}$	99.72 ± 0.02	99.76 ± 0.03	99.68 ± 0.04	99.67 ± 0.05	99.60 ± 0.04

the spatial size of 9×9 as the input size of the proposed framework.

4) Impact of Diverse Training Percentage: The percentage of training data plays an important impact on the experimental accuracies. For this point, the following experiments are done under different percentages on training samples for the three data sets. We take 5%, 10%, 15%, 20%, and 25% of the available data as training samples for the IN and KSC data sets and record the classification results, as shown in Table IX. Similarly, we experiment with 1%, 2.5%, 5%, 7.5%, 10%, and 12.5% of data as the training set for the UP data set. The results are listed in Table X. As can be seen, the classification accuracies increase rapidly and then grow slowly with the increase of training percentage. In particular, the test accuracies vary little when the training percentage is more than 20% and 10% for the three data sets.

5) Impact of Hyperparameter r: Actually, the hyperparameter r determines the number of neurons in the FC layer. To this end, set r from 1 to 8 empirically to discuss the impact of different r values in terms of OAs. The results are listed in Table XI. According to the experimental observation, we empirically select the appropriate $r = 4$ for the IN and KSC data sets and $r = 3$ for the UP data set based on the best OAs.

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IV. Conclusion

This article develops an FGN with SSCA strategy for HSI classification problem. The main idea is to concentrate on significant spectral bands and spatial locations by SSCA module from the raw input in a continuous manner, and then, the high level of spectral–spatial features is learned by FGN, which could fully utilize the information between the bands and obtain more available and valuable features. The fusion of these different features enhances the information flow between features. This is beneficial to the HSI classification and greatly improves generalization performance. The experimental results on several HSI data sets support the feasibility and effectiveness of the proposed FG-SSCA method and exhibit better visualization images compared with the state-of-the-art approaches for HSI classification.

However, there is still room to further study in the future. For example, it is an interesting problem to investigate the statistical significance of different methods to further evaluate the performance of the proposed method. In addition, we will also explore the issue of sample scarcity [55], which is challenging. Next, transfer learning [56] and unsupervised training methods [57] may be integrated into our framework to alleviate the aforementioned problem. Also, it is expected that the low-rank subspace clustering [58] could combine with the deep learning method to enhance the prior information for improving HSI classification performance.

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